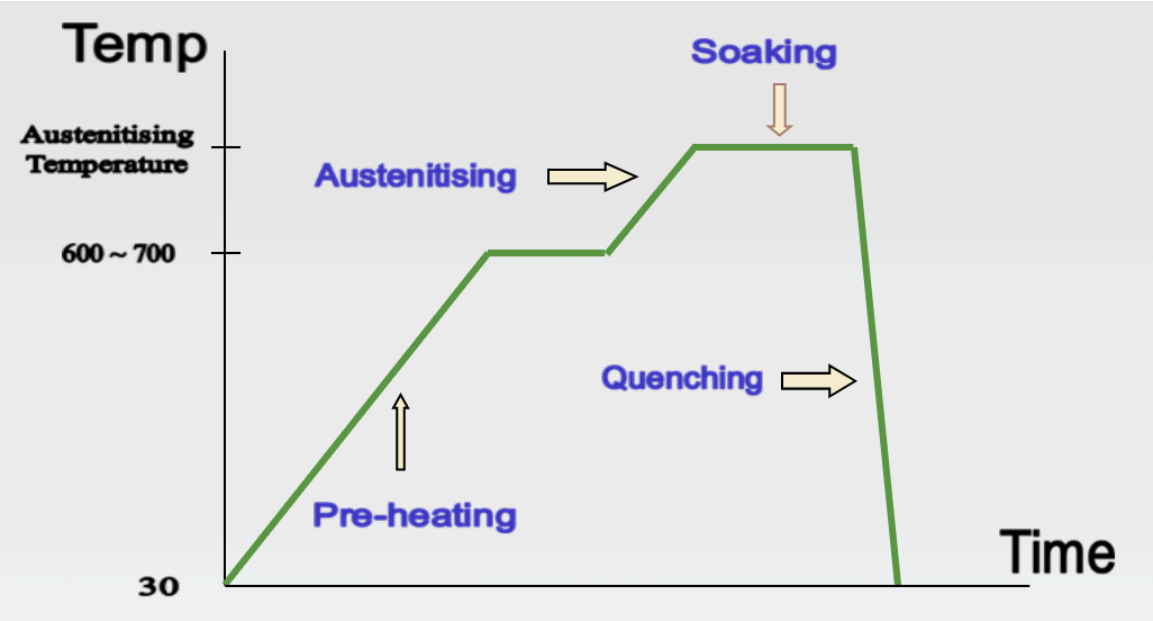
Heat Treatment of Steels

Recall the basic premise of heat treatment, involving heating to a temperature, holding it, and quenching.

For steels however, the process involves 4 steps:



Pre heating and Austenitising is done first to reach the austenitising temperature. If normal heating is done without any pre-heating, the heating rate of austenitising may be too fast and for highly stressed steels, distortion or cracking may occur. Instead, a separate furnace is used to heat the steel at a slower rate to a higher temp, before austenitising is done at a higher rate (due to economical reasons) to reach the holding temp.

Soaking is done to ensure the whole component has been uniformly heated to a consistent austenitic microstructure.

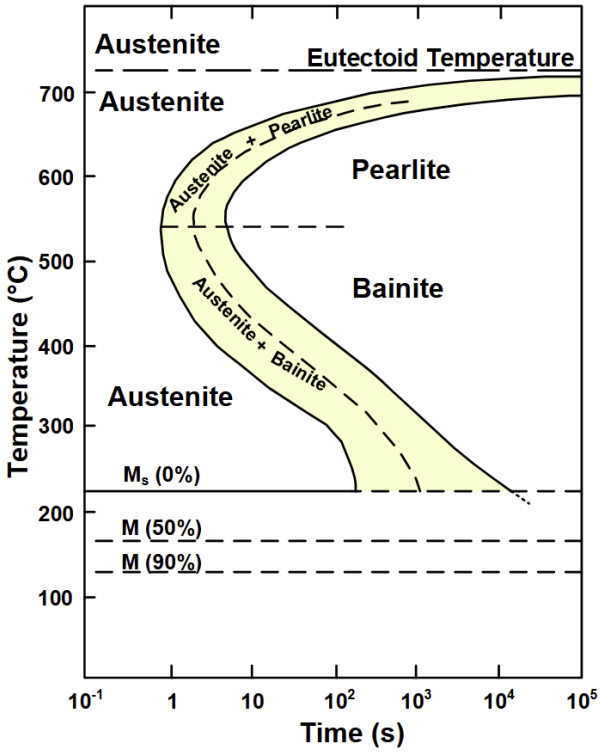
Quenching is done last by using an appropriate quenching medium to obtain a specific cooling rate.

Time-Temperature-Transformation (TTT) Diagram

The standard Iron-Carbon Equilibrium phase diagram only shows phases under equilibrium, and does not include phases like martensite which are

formed under rapid quenching. The TTT diagram is a solution to this, as it accounts for the elapsed time.

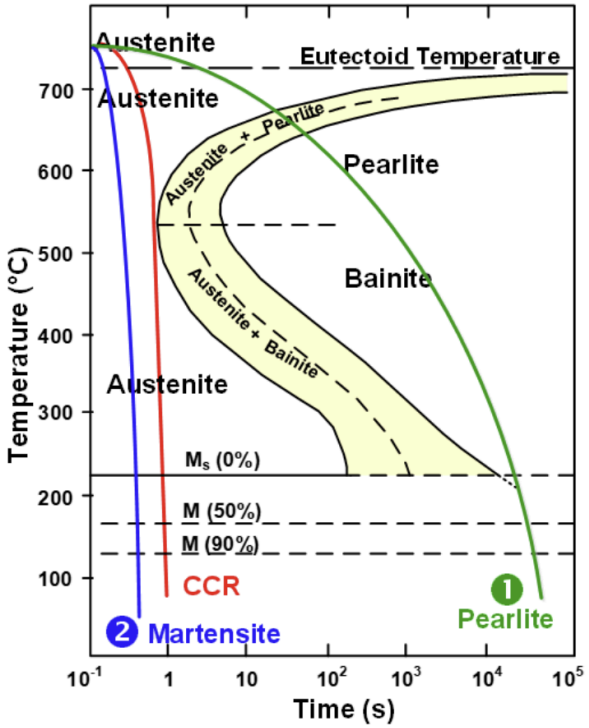
The TTT diagram is mostly used for soaking and quenching, as that is when the time elapsed matters the most.



Essentially, the TTT diagram shows the different microstructures and phases it goes through depending on the time and temperature, for a given steel composition. The one shown above describes the different microstructures of eutectoid steel.

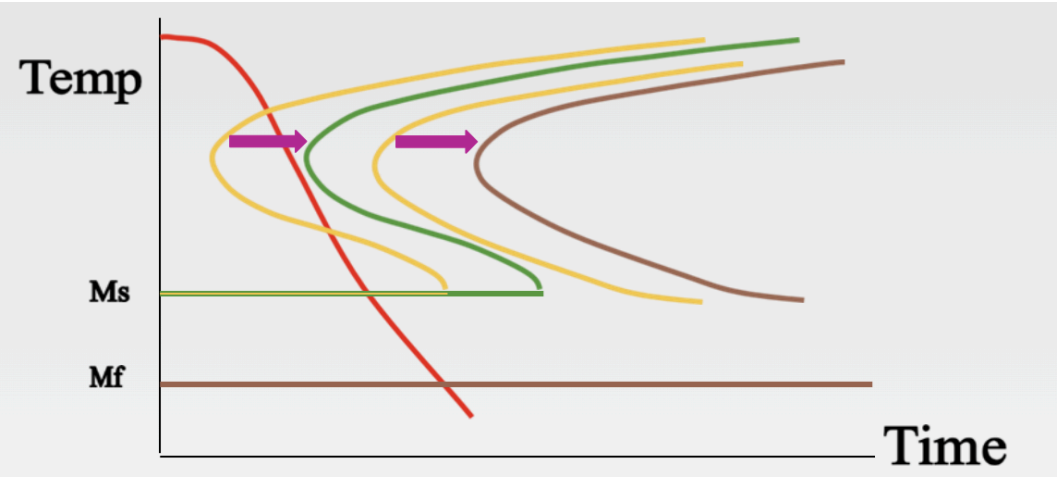
In the example of hardening, say the metal starts at the austeniting temperature and is quenched. To be properly hardened, the path on the TTT diagram is represented by a decreasing line that stays in the austenite region, thus allowing the microstructure to stay as austenite until it passes the martensite line, M_s. Notice how this requires the line to be steep enough to fall behind the maximum point of the curve.

If the cooling rate was not steep enough, it would cross into the pearlite region and when it cooled below the martensite line, martensite will not be formed.



Factors Influencing the TTT diagram

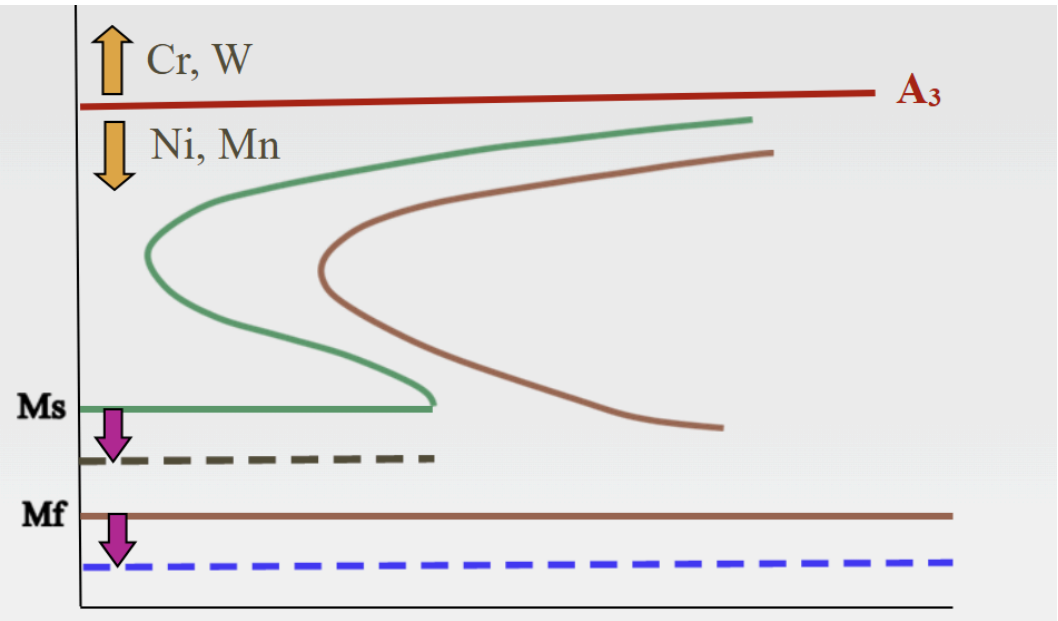
Adding alloying elements (other than Co) essentially shift the curve to the right, providing more clearance and decreasing the CCR, thus providing easier hardenability.



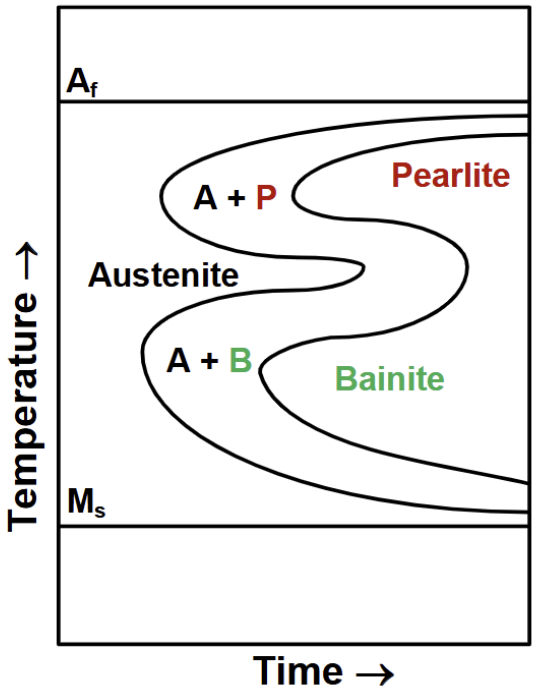
When alloying elements that form carbides are added, e.g. Cr and W, the austeniting temperature is raised, which can be thought of as harder steels (those with carbides) being more difficult to further harden, shown by the increase the austeniting temperature.

Elements that dissolve in ferrite like Ni and Mn lower the austeniting temperature instead, which can be thought of as softer/tougher steels being able to be hardened much more/easily due to the lower austeniting temperature.

However, adding any alloying elements cause the martensite lines to decrease/martensite temperatures to drop.



In some cases, the TTT curve may change completely to have two noses, one for pearlite and one for bainite. The protrusion for bainite makes it possible to cool from the furnace to room temp to achieve bainite; impossible for plain carbon steels.

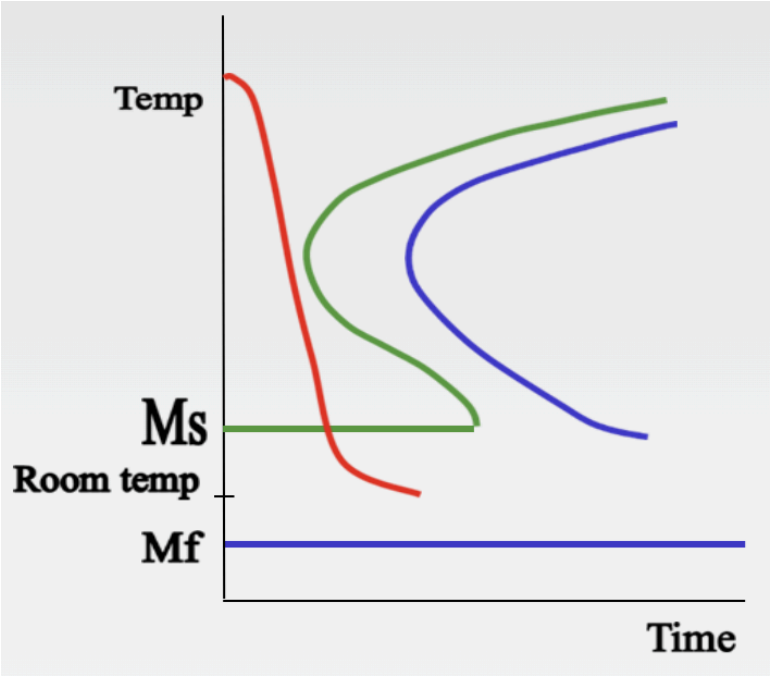


Elements like W, V, Ti and Mb form very hard and complex carbides, which are difficult to dissolve in austenite and thus raise the austeniting temperature.

Retained Austenite

One issue that may occur with heat treatment is retained austenite.

For steels with high amounts of carbon or other alloying elements, the martensite temperatures, specifically Mf may drop well below room temperature. This prevents normal quenching methods from transforming all of the austenite to martensite, causing austenite to be dissolved in the martensite, thus having some austenite retained.



Retained austenite causes a drop in the overall hardness (due to the dissolved softer austenite), may cause spontaneous cracking if the steel happens to worked below the room temp (as the austenite may spontaneously form into martensite) and is dimensionally unstable due to defects caused by spontaneous martensite formation.

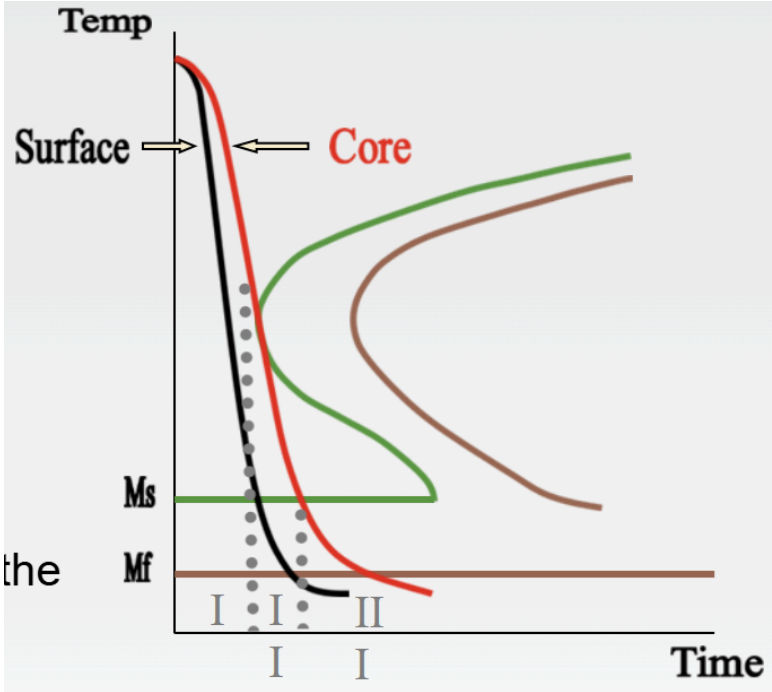
To eliminate retained austenite, there are two methods,

Double tempering; by tempering once to temper the original martensite and transform the austenite to martensite, and tempering again to temper the newly formed martensite.

Sub Zero Treatment; by refrigerating to as cold as -98C to ensure a consistent martensite microstructure, before being warmed back to room temperature.

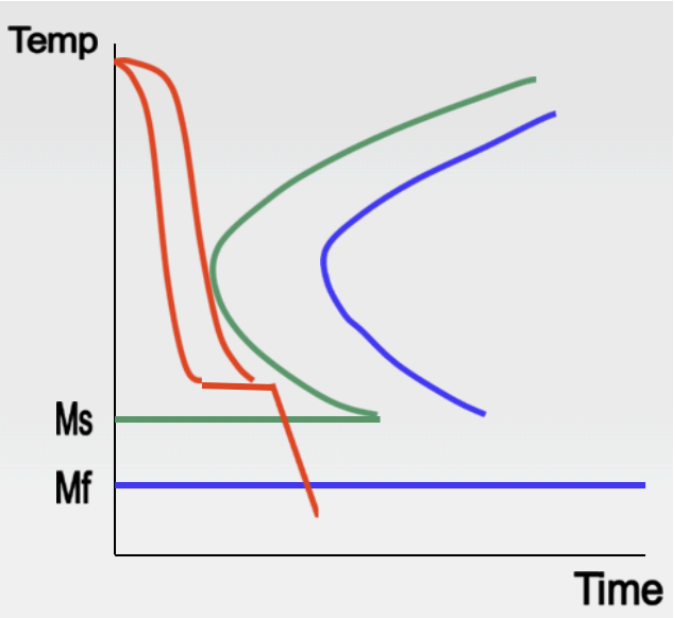
Distortions & Cracking

Recall the size effect from heat treatment, where the surface may cool much faster than the core. More specifically, this is due to the different heating rates causing the surface to transform into martensite earlier than the core. Since austenite is denser than martensite, there will be some expansion which causes distortions and cracking.



To fix distortions and cracking via remedial measures, we can

- hang long and thin steels vertically in the furnace/ ensuring the material is well supported
- minimise stress raisers like sudden section changes, sharp corners, rough surfaces, thin sections etc.
- minimise the cooling rate as much as possible with a suitable medium
- ensure an appropriate steel is picked for the design, e.g. air quenched steels
- martempering heat treatment may be done to minimise the size effect.
    - Martempering/Marquenching involves quenching in a molten salt or metal bath kept at a temperature above Ms. This means that the metal will equalise to a temperature just above Ms, and the less dense martensite will not form yet, preventing cracking and distortions as the microstructure is still austenite.
    - Then, the metal is air or oil cooled to fully form the martensite, before being tempered.



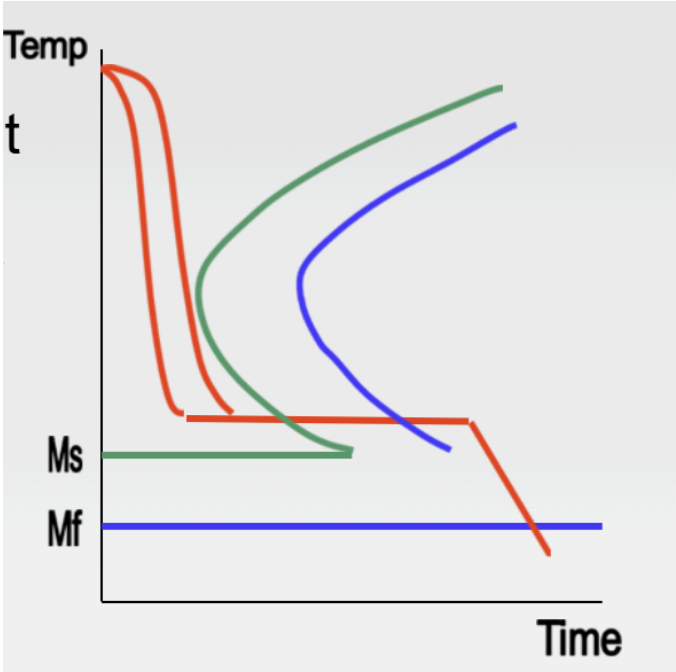
One restriction of this method is that it is only applicable to smaller alloy steel components

Austempering/Bainitic Hardening

Similar to martempering, there is austempering, also called Bainitic Hardening.

Again, the component is quenched similarly to right above the Ms temperature, but is held there for a sufficient amount of time to allow the austenite to transform into bainite.

Then, the part is air-cooled to form bainite.



Again, this is mainly limited to smaller components.

Protective Atmosphere in Heat Treatment

If the surface of the metal is not protected, reactions with the furnace environment and the metal surface may occur, leading to negative effects with the

Decarburisation

Occurring in medium and high carbon steels, this reaction occurs when the carbon in the steel reacts with the CO_2 in the furnace, causing CO to form outside the steel. This causes a drop in C in the metal surface, leading to lower surface hardness, wear & fatigue resistance.

To avoid this, the ratio of CO/CO_2 should be carefully controlled, or the surface protected from reacting.

Scaling

Scaling is the oxidation of the metal, either due to O_2, CO_2 or water vapour, most commonly occurring in low carbon steels.

This can be mitigated similarly to decarburisation, but more control is needed to avoid carbon pickup, which changes the %C of the low carbon steel.

Surface Heat Treatment

Surface heat treatment is done on components where wear and fatigue are much more important factors, compared to tensile strength. These components should undergo surface heat treatment instead of complete hardening, as the surface is more critical.

The main objective of surface heat treatments is to form a strong and hard surface and a soft but tougher core. To obtain this, a strong and tough steel is first picked to undergo surface heat treatment, and the surface is heat treated to optimise wear resistance, heat resistance and other similar characteristics.

There are two main ways to do surface heat treatment, diffusion via thermochemical processes and selective hardening.

Diffusion Methods

Generally, these methods involve modifying the surface's chemical composition by introducing carbon, nitrogen, or other elements.

The main working principle is that the element must be soluble in the steel at the specific temperature. This causes the surface volume to increase, introducing compressive stresses at the surface which increase fatigue resistance

Generally, the two types of diffusion are,

Ferritic, at temps <600C, where less C or N is dissolved, causing lesser distortion,

Austenitic, at temps >800C, where more C or N can be dissolved, at the expense of more distortion.

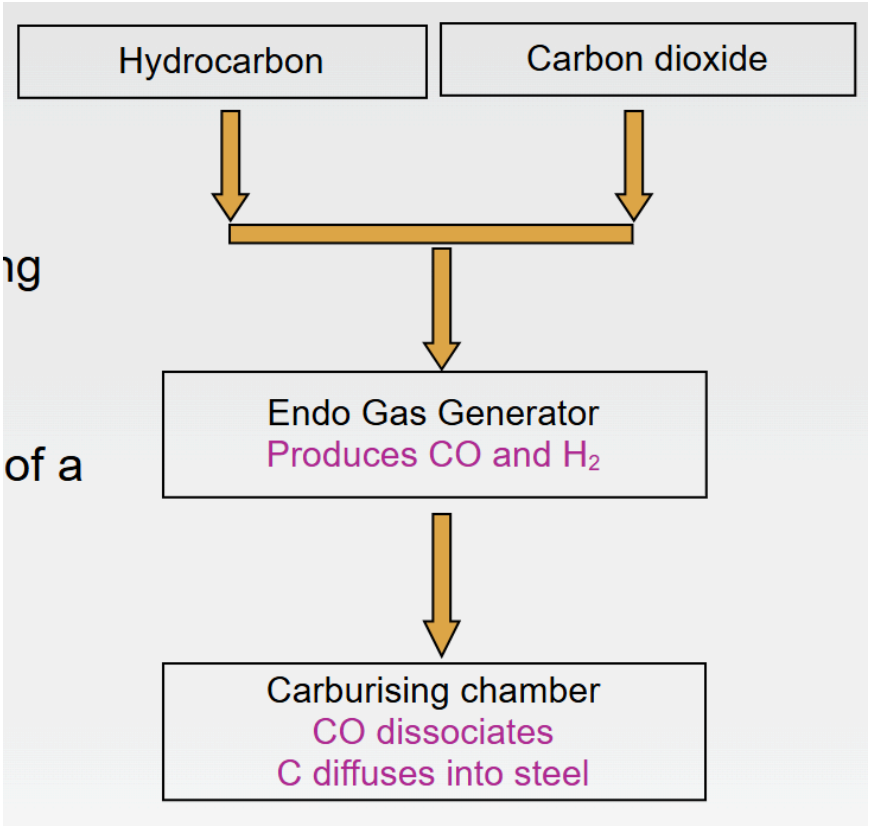
There are 3 main diffusion methods:

Carburising

Carburising is done on steels with C < 0.2%, and is meant to increase %C on surface to form a hard martensitic case (with the case increased to >0.8% C)

Steel is heated to above 880C (usually around >925C) while in contact with carbonaceous materials, causing the carbon to be diffused into the steel surface, producing a case of higher carbon content. The steel is then quenched accordingly to form a martensitic case.

The most practical form of carburising is gaseous carburising, where the carbonaceous material introduced is a gas. This is most suitable for large quantities of small components. Components are heated to the carburising temperature and are held for 3 to 4 hours in a controlled atmosphere, with hydrocarbon gases like CH4 and C3H8 and carbon dioxide. This produces CO as the two gases react in the endo gas generator. The CO then disassociates at the surface of the component and diffused C into the steel.



Advantages:

- Clean; little housekeeping needed compared to solid/liquid carburising
- %C of surface can be more accurately and easily controlled

Disadvantages:

- Complex furnaces and equipment needed
- Intricate process controlled required

Nitriding

Nitriding is used for hardening finish-machined metals and heat treated steels. or The most common method, gaseous nitriding, involves heating the component at 500C for 2 to 4 days in an ammonia circulated atmosphere

This causes the ammonia to dissociate at the surface and nitrogen is diffused into the steel surface. As the steel is being treated in the ferritic stage, no quenching is needed, thus minimising distortions.

Typically, this process is not useful for plain carbon steels, as the iron nitrides formed from the diffusion will go to a higher depth, causing a lower surface hardness as less of it is concentrated at the center. A brittle white layer also forms if plain carbon steels are used which is not ideal for mating surfaces.

Instead, nitralloys should be used; alloys with Al, Cr and Mo and ~0.3 to 0.5%C. Usually, due to the AlN formed which restricts diffusion, a 0.2 to 0.4mm case is formed, without a need for further machining.

Advantages:

- low processing temps used; less distortion
- hard case obtained (HRC > 70)
- good corrosion resistance
- clean and simple operation

Disadvantages:

- temp cannot exceed 500 otherwise hardness will be lost
- toughness reduced
- if too much N introduced, white layer forms

Carbonitriding

Carbonitriding is used for mild and low carbon alloy steels/plain carbon steels. It involves introducing carburising gas and ammonia to introduce both C and N into the alloy steel while heated to 820-840C for 3 hours, before quenching in oil. The mixture of the gases can have their proportions varied

Advantages:

introducing N stabilises austenite, slower transformation and less distortion

Selective Surface Hardening

The general idea of selective surface hardening, or selective surface heat treatment, is that heat is applied locally to only transform the surface but not the core.

Basically, SSH does not cause any composition change unlike diffusion, and the surface will have the same properties as being heat treated traditionally.

To apply head locally, rapid heat must be applied to austenitise the surface, before quenching it, while preventing the core from transforming. The higher the heating intensity, the shallower the case.

Usually, we use SSH for larger components or for hardening only a certain section of the part. The steel used should be 0.3-0.6 %C for austenite and thus martensite to form.

Flame Hardening

A selected area is heated with a flame above the UCT, usually ~800C, before being immediately quenched with cold water, giving a surface hardness of 50 to 60 HRC This is for 0.4% to 0.95%C and low alloy steels. This process is the easiest and cheapest especially for larger components.

To control the depth of the case, we can control the gas mixture used (oxygen- acetylene), influencing the heating rate and thus the case thickness, or by increasing the heating rate by bringing the burner closer to the component, which causes a thicker case. (?????????????)

Induction Heating

An inductor coil has high frequency AC passed through, introducing eddy currents which raise the component's temperature in the inducted area. It is then quenched with pressure jets of water. Medium carbon or low carbon

faster

no oxidation & carbodisation????

minimal distortion

easy control of case depth

costly equipment

Generally, the heating area, or the heating pattern is infleuced by

the shape and number of coils,

operating frequency

power input,

workpiece nature

but generally, the higher the operating frequency, the thinner the case.